EXPVID: A BENCHMARK FOR EXPERIMENT VIDEO UNDERSTANDING & REASONING

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ABSTRACT

Existing multimodal large language model (MLLM) benchmarks primarily focus on generic actions or knowledge tasks but often overlook the nuanced, long-horizon nature of authentic scientific experimentation. To address this gap, we introduce ExpVid, the first benchmark specifically designed for evaluating MLLMs on experiment video understanding and scientific reasoning. ExpVidcomprises 7,800 question-answer pairs derived from 390 peer-reviewed laboratory videos spanning 13 scientific disciplines, with an average duration of 489 seconds per video. Our benchmark features a novel three-level hierarchy: Level-1 evaluates fine-grained visual perception (tool, material, quantity, and operation recognition), Level-2 assesses procedural understanding (step ordering, sequence generation, completeness verification, and step prediction), and Level-3 tests scientific reasoning capabilities (experimental analysis and discovery). Through vision-centric annotation design, ExpVidprevents models from relying on textual shortcuts and ensures genuine visual comprehension. Comprehensive evaluation of 19 state-of-the-art MLLMs reveals significant performance gaps, with the best model (GPT-5) achieving only 57.5% on Level-2 and 56.4% on Level-3 tasks. Our analysis demonstrates the critical importance of visual information and identifies key challenges in procedural prediction and long-horizon reasoning, providing a rigorous evaluation framework for developing MLLMs capable of understanding real-world scientific experiments.

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1 Introduction

The rapid advancement of multimodal large language models (MLLMs) has led to remarkable progress in understanding and reasoning about visual content. However, despite significant strides in general video understanding [Li et al., 2024, Fu et al., 2025], existing benchmarks predominantly focus on everyday activities, entertainment content, or simplified educational scenarios. This leaves a critical gap in evaluating MLLMs' ability to comprehend the sophisticated, methodical nature of authentic scientific experiments, which represent one of humanity's most systematic approaches to knowledge discovery.

Scientific experimentation videos present unique challenges that distinguish them from conventional video understanding tasks. Unlike general videos that may rely on common sense or everyday knowledge, experiment videos require *precision* in recognizing specialized tools and materials, *procedural understanding* of multi-step protocols, and *scientific reasoning* to connect observations with underlying principles. Furthermore, authentic laboratory videos often span extended durations with subtle but critical visual details that determine experimental success or failure.

Consider a molecular biology experiment for protein purification: an MLLM must not only identify specific laboratory equipment (centrifuges, pipettes, gel electrophoresis apparatus) and reagents (buffers, enzymes, molecular weight markers), but also understand the sequential dependencies between steps (cell lysis must precede protein extraction), recognize quantitative precision (exact volumes and concentrations), and ultimately reason about the experimental outcomes in relation to scientific principles. Such capabilities are essential for MLLMs to serve as effective scientific assistants or educational tools.

To address this gap, we introduce **ExpVid**, the first comprehensive benchmark designed specifically for evaluating MLLMs on experiment video understanding and scientific reasoning. ExpVidis built upon 390 carefully curated peer-reviewed laboratory videos from the Journal of Visualized Experiments (JoVE) [JoVE], spanning 13 scientific disciplines including molecular biology, chemistry, neuroscience, and materials science. Our benchmark encompasses 7,800 expertly annotated question-answer pairs organized across three hierarchical levels that mirror the scientific workflow: from fine-grained perception to procedural understanding to high-order reasoning.

Our key insight is that scientific experiment understanding requires a *three-level hierarchy* spanning perception, procedure, and reasoning, with *vision-centric* annotations that prevent shortcuts via textual priors. Level-1 tasks evaluate fine-grained visual perception through multiple-choice questions on tool recognition, material identification, quantity estimation, and operation classification. Level-2 tasks assess procedural understanding via step ordering, sequence generation, completeness verification, and step prediction. Level-3 tasks test scientific reasoning through experimental analysis and discovery using fill-in-the-blank questions that require synthesizing information across entire experiments.

We evaluate 19 state-of-the-art MLLMs, including both open-source and closed-source models, revealing substantial performance gaps across all levels. The best-performing model, GPT-5, achieves 57.5% accuracy on Level-2 and 56.4% on Level-3 tasks, while the top open-source model (InternVL3-78B) reaches only 41.9% and 37.7% respectively. Notably, human performance baselines show that Level-1 tasks are particularly challenging even for humans (37.6%), highlighting the precision required for scientific visual understanding.

Our contributions are fourfold: (1) We introduce the first benchmark specifically targeting authentic experiment video understanding with a hierarchical evaluation framework; (2) We propose a vision-centric, semi-automatic annotation pipeline that combines LLM assistance with expert validation to ensure high-quality, bias-free questions; (3) We curate a comprehensive dataset of 390 peer-reviewed experiments across 13 disciplines with multi-scale temporal annotations; and (4) We provide extensive evaluation of 19 MLLMs, identifying critical gaps in procedural prediction and long-horizon scientific reasoning that can guide future model development.

2 Related Work

Video Understanding Benchmarks. The landscape of video understanding benchmarks has evolved from action recognition [**?**] to complex reasoning tasks. Recent works like MVBench [Li et al., 2024] and Video-MME [Fu et al., 2025] provide comprehensive evaluations across multiple video understanding tasks, while benchmarks like MLVU [Zhou et al., 2024] and VRBench [Yu et al., 2025] focus on long-form video comprehension and reasoning. However, these benchmarks primarily target general-domain videos with everyday activities or narrative content, lacking the specialized vocabulary and precise procedural understanding required for scientific contexts.

Scientific and Knowledge-driven Benchmarks. Several benchmarks evaluate MLLMs on scientific knowledge and reasoning. MMMU [Yue et al., 2024] and Video-MMMU [Hu et al., 2025] assess understanding across multiple academic disciplines, while domain-specific benchmarks like MathVista [Lu et al., 2023] and ScienceQA [Saikh et al., 2022] focus on mathematical and scientific question answering. SciBench [Wang et al., 2023] evaluates scientific problem-solving capabilities, and the recent SFE benchmark [Zhou et al., 2025] examines scientific figure understanding. However, these benchmarks primarily rely on static images or textual content, missing the dynamic, temporal nature of actual experimental procedures.

Multimodal Large Language Models. The development of MLLMs has progressed from early vision-language models [**?**] to sophisticated architectures capable of processing multiple modalities [**??**]. Recent models like GPT-4V [**?**], Gemini [**?**], and open-source alternatives such as InternVL [**?**] demonstrate strong performance across various vision-language tasks. However, their evaluation on authentic scientific content, particularly dynamic experimental videos, remains limited.

Educational Video Understanding. Some works have explored educational content analysis, focusing on lecture videos [**?**] or instructional content [**?**]. However, these approaches typically address general educational scenarios rather than the specialized, precision-critical domain of scientific experimentation, where subtle visual details and exact procedural sequences are paramount.

Our work differs fundamentally from existing benchmarks by specifically targeting authentic scientific experiment videos with a hierarchical evaluation framework that spans from fine-grained perception to scientific reasoning. Unlike general video benchmarks that rely on common knowledge, ExpVidrequires specialized scientific understanding, procedural precision, and the ability to reason about complex experimental outcomes. Our vision-centric design also prevents models from exploiting textual shortcuts, ensuring genuine visual comprehension capabilities are evaluated.

3 ExpVid: A Scientific Experiment Video Benchmark

3.1 Design Principles

ExpVidis designed around three core principles that distinguish it from existing video understanding benchmarks:

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Authenticity. Rather than using simplified or staged educational content, ExpVidexclusively utilizes peer-reviewed experimental videos from JoVE [JoVE], ensuring that evaluation reflects real scientific practices and standards. These videos undergo rigorous peer review and represent established experimental protocols across diverse scientific disciplines.

Hierarchical Evaluation. Scientific understanding progresses from basic perception to complex reasoning. Our three-level hierarchy mirrors this progression: Level-1 evaluates fine-grained visual recognition capabilities, Level-2 assesses procedural understanding of experimental workflows, and Level-3 tests high-order scientific reasoning and discovery.

Vision-Centric Design. To prevent models from exploiting textual shortcuts present in transcripts or captions, our annotations are crafted to require visual information for correct answers. Questions are designed such that they cannot be answered from audio transcripts or general scientific knowledge alone.

3.2 Three-Level Task Hierarchy

Figure 1 illustrates our comprehensive task taxonomy across three hierarchical levels:

Level-1: Fine-grained Perception. This level evaluates basic visual recognition capabilities essential for scientific observation:

- *Tool Recognition*: Identifying specialized laboratory equipment (microscopes, centrifuges, pipettes)
- *Material Recognition*: Recognizing reagents, samples, and experimental materials
- *Quantity Recognition*: Estimating volumes, concentrations, and numerical measurements
- *Operation Recognition*: Classifying specific experimental actions and manipulations

All Level-1 tasks use multiple-choice format with carefully crafted distractors. For tools and materials, distractors include visually similar or functionally related items. For quantities, near-range numerical options test precision in measurement reading. For operations, confusable actions that appear similar but have different purposes serve as distractors.

Level-2: Procedural Understanding. This level assesses comprehension of experimental workflows and temporal dependencies:

- *Step Ordering*: Arranging experimental steps in correct temporal sequence
- *Sequence Generation*: Producing complete step-by-step experimental protocols
- *Completeness Verification*: Identifying missing or incorrect steps in protocols
- *Step Prediction*: Predicting the next logical step in an experimental sequence

Level-2 tasks require understanding causal relationships and procedural dependencies. Evaluation metrics include Top-1 accuracy for ordering tasks and Jaccard similarity for sequence generation tasks.

Level-3: Scientific Reasoning. This level tests high-order reasoning and scientific discovery capabilities:

- *Experimental Analysis*: Analyzing experimental design, controls, and validity
- *Scientific Discovery*: Drawing conclusions and connecting results to scientific principles

Level-3 uses cloze-style questions requiring synthesis of information across entire experiments. Evaluation employs LLM-based judging (Phi-3-mini) with per-blank accuracy assessment.

3.3 Dataset Construction Pipeline

Figure 2 provides an overview of our systematic construction process involving collection, annotation, and verification stages:

Data Collection and Curation. We begin with JoVE's extensive repository of peer-reviewed experimental videos, paired with detailed research papers and ASR transcripts. Our selection process involves:

1. *Automated Filtering*: Using DeepSeek-R1, we evaluate videos on five criteria—Continuity, Alignment, Clarity, Integrity, and Focus—requiring overall scores $\geq$4.0 and no individual criterion $<$3.5.
2. *Duration Normalization*: Videos are trimmed to the 25th–75th percentile range (378–728 seconds) to ensure consistent temporal scope.

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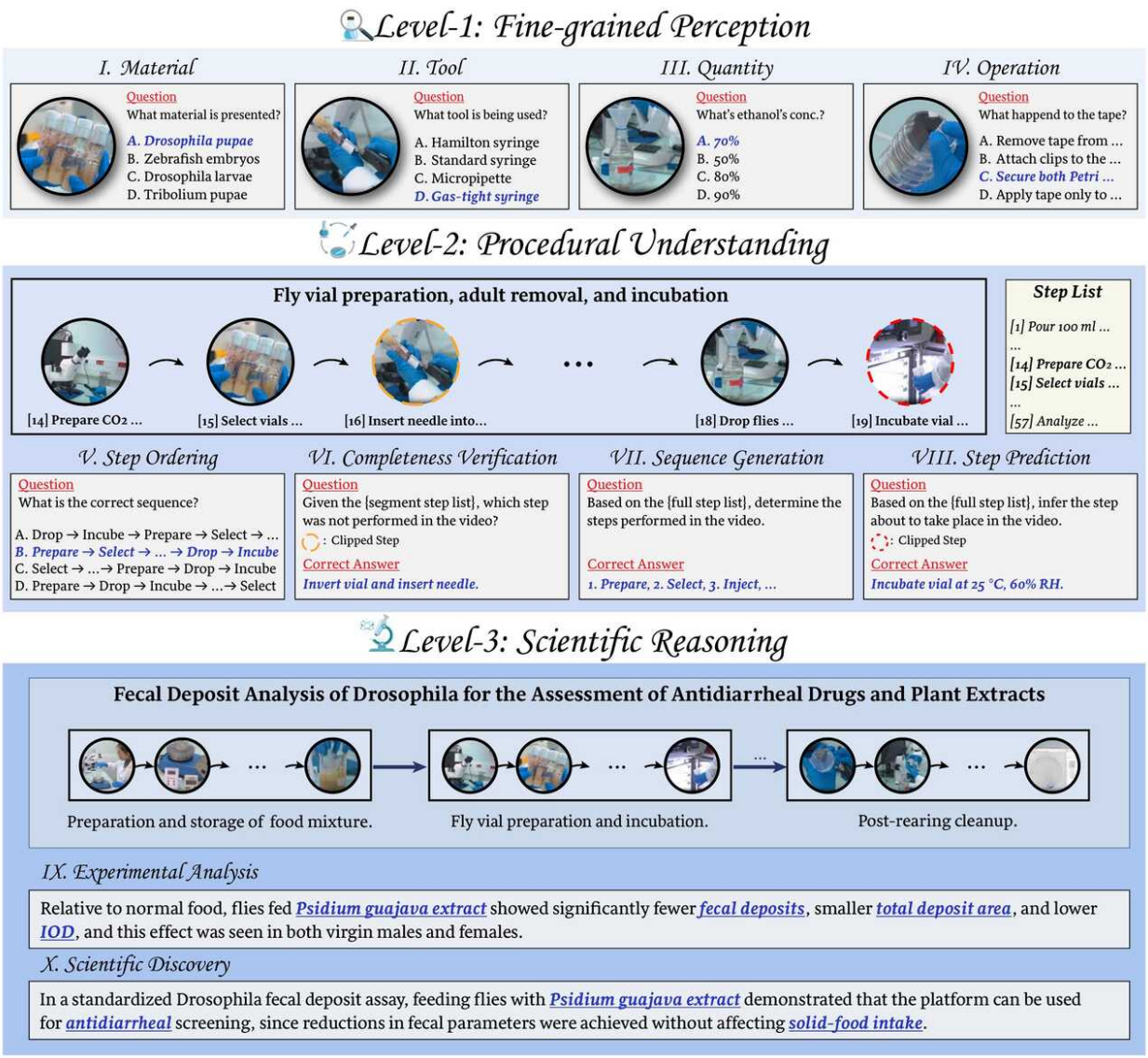

Figure 1: Illustration of the three-level task hierarchy in ExpVid. Level-1 focuses on fine-grained perception, Level-2 on procedural understanding, and Level-3 on scientific reasoning.

3. *Expert Curation*: PhD-level researchers in biology, chemistry, and medicine select 30 high-quality experiments per discipline, resulting in 390 videos across 13 scientific domains.

Multi-Level Preprocessing. Videos are processed differently for each level:

- *Level-1*: Videos are segmented by ASR sentence boundaries, creating approximately 10,000 action-level clips averaging 8 seconds each.
- *Level-2*: Stage boundaries are identified using DeepSeek-R1, producing approximately 3,500 stage-level segments averaging 48 seconds each.
- *Level-3*: Full experimental procedures are retained, with concluding slides and data analysis sections removed to prevent textual shortcuts.

Vision-Centric Annotation. Our annotation process ensures questions require visual understanding:

- *Level-1*: Targets are extracted using DeepSeek-R1, verified for visual presence using Qwen2.5-VL captions, and formulated into 4-option MCQs with carefully designed distractors.

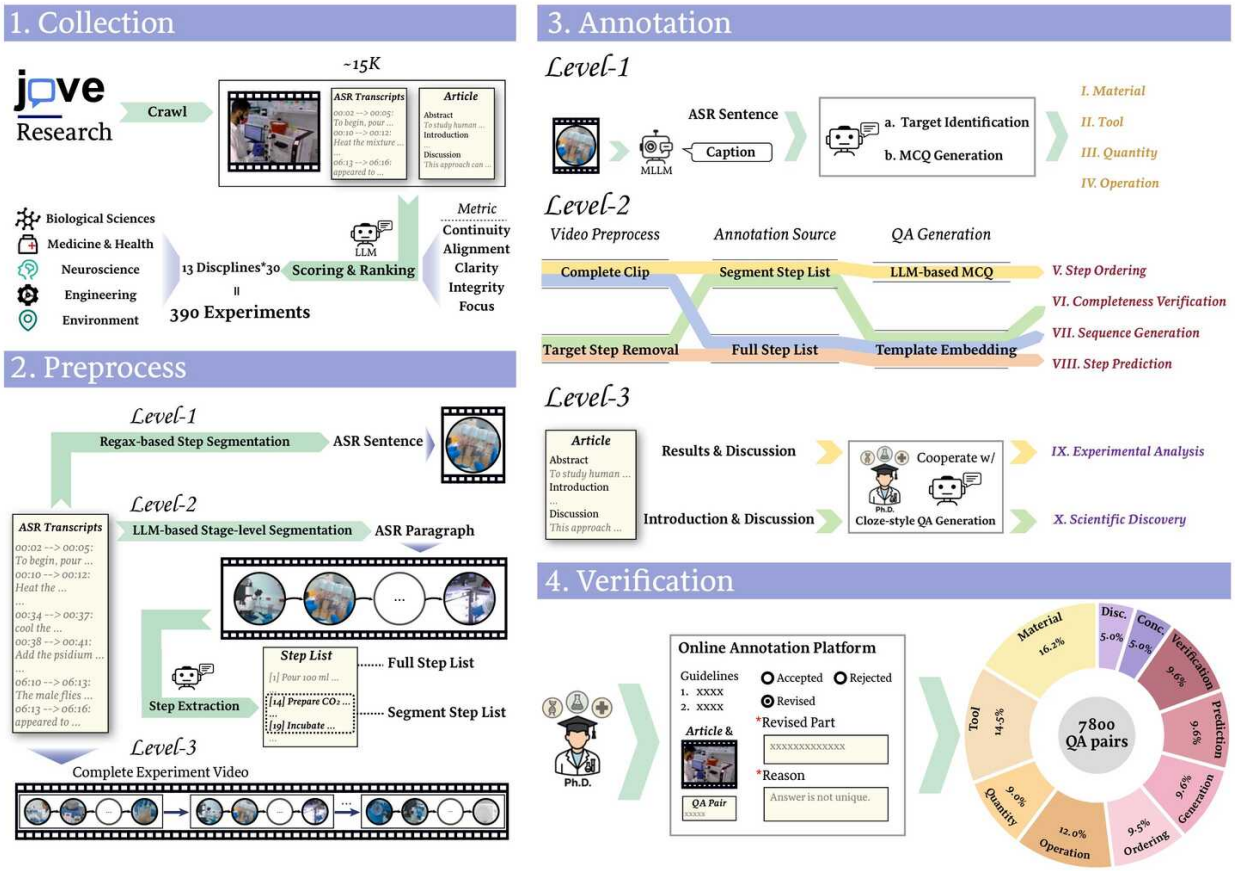

Figure 2: Overview of the ExpVidconstruction pipeline, including collection, multi-level annotation, and expert verification stages.

- *Level-2*: Step lists are constructed from procedural descriptions, then used to generate ordering challenges, candidate sequences, missing-step detection, and prediction tasks.
- *Level-3*: Relevant paper sections are extracted using MinerU, summarized by GPT-5, and PhD experts craft cloze questions requiring full-video reasoning that cannot be answered from text alone.

Expert Verification and Quality Control. All annotations undergo validation by PhD-level domain experts who verify:

- Visual clarity and detectability of target elements
- Plausibility and appropriateness of distractors
- Alignment between step descriptions and video segments
- Uniqueness and unambiguity of correct answers

Expert annotators spend an average of 0.3 hours per Level-1 question, 0.5 hours per Level-2 question, and 1.2 hours per Level-3 question, reflecting the increasing complexity and precision required across levels.

3.4 Dataset Statistics

ExpVidcomprises 7,800 verified QA pairs distributed across 10 tasks spanning three hierarchical levels. The dataset encompasses 390 experimental videos averaging 489 seconds in duration, covering 13 scientific disciplines including Molecular Biology, Biochemistry, Neuroscience, Materials Science, and Chemistry. Table 1 positions ExpVidwithin the landscape of existing MLLM benchmarks, highlighting its unique focus on authentic scientific content with both automatic and manual annotation processes.

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Table 1: Comparison between ExpVid and existing MLLM benchmarks. A and M indicate automatic and manual annotation, respectively.

Benchmark	#QA Pairs	#Videos	Avg. Sec.	#Task Types	Annotation	Domain
General Video Benchmarks						
MVBench [Li et al., 2024]	4,000	3,641	16.0	20	A+M	General
Video-MME [Fu et al., 2025]	2,700	900	1,017.9	1	M	General
MLVU [Zhou et al., 2024]	3,102	1,730	930.0	9	M	Narrative
VRBench [Yu et al., 2025]	9,468	960	5,796.0	1	M	Narrative
Knowledge-driven Benchmarks						
MMVU [Zhao et al., 2025]	3,000	1,529	51.4	2	M	Multi-disc.
Video-MMMU [Hu et al., 2025]	900	300	506.2	3	M	Multi-disc.
MathVision [Wang et al., 2024]	3,040	–	–	1	M	Math
MathVista [Lu et al., 2023]	6,141	–	–	31	M	Math
MMMU [Yue et al., 2024]	11,500	–	–	2	M	Multi-disc.
ScienceQA [Saikh et al., 2022]	21,208	–	–	1	M	Science
SciBench [Wang et al., 2023]	789	–	–	1	M	Science
MMStar [Chen et al., 2024]	1,500	–	–	6	M	Multi-disc.
SFE [Zhou et al., 2025]	830	–	–	66	M	Science
ExpVid	7,800	390	489.0	10	A+M	Science

4 Experiments

4.1 Experimental Setup

Model Selection. We evaluate 19 state-of-the-art MLLMs spanning both open-source and closed-source categories. Open-source models include the Qwen2.5-VL series, InternVL3 variants, GLM-4V models, and specialized thinking models like MiMo-VL and Kimi-VL. Closed-source models encompass GPT-5, Gemini-2.5-Flash/Pro, Claude-Sonnet-4, and Seed-VL. For models supporting chain-of-thought reasoning, we evaluate both standard and "thinking" modes.

Evaluation Metrics. We employ task-appropriate evaluation metrics: Top-1 accuracy for multiple-choice questions (Level-1), Jaccard similarity for sequence generation tasks (Level-2), and LLM-based judging using Phi-3-mini with per-blank accuracy for cloze questions (Level-3). Human performance baselines are established through expert annotation of a subset of questions.

Implementation Details. Videos are uniformly sampled to extract frames for model input. We use standard prompt templates across all models to ensure fair comparison, with task-specific instructions for each level. For models with context length limitations, we employ adaptive frame sampling strategies while maintaining temporal coverage of the entire experimental procedure.

4.2 Main Results

Table 2 presents comprehensive performance results across all 19 evaluated models and 10 tasks. Several key findings emerge from this evaluation:

Performance Hierarchy. Closed-source models generally outperform open-source alternatives, with GPT-5 achieving the highest overall performance (57.5% Level–2, 56.4% Level–3). Among open-source models, InternVL3–78B shows the strongest performance (50.9% Level–1, 41.9% Level–2, 37.7% Level–3), demonstrating the benefits of scale in scientific video understanding.

Level-specific Insights. Level–1 tasks prove challenging even for humans (37.6% baseline), highlighting the precision required for scientific visual recognition. Interestingly, some models surpass human performance on specific Level–1 tasks, particularly quantity recognition where InternVL3–78B achieves 73.2%. Level–2 shows the largest performance gaps, with GPT-5 reaching 57.5% while many models struggle below 30%. Level–3 tasks demonstrate the most consistent challenge across all models, with the best performance at 56.4%.

Task-specific Performance. Within Level–2, models show striking asymmetry: Step Ordering achieves relatively high accuracy (InternVL3–78B: 87.1%), while Step Prediction proves extremely challenging (many models near 0%).

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Table 2: Performance of evaluated models on ExpVidacross 10 tasks under three levels.

Model	Think	Level-1					Level-2					Level-3		
		Tool	Mat.	Quan.	Oper.	Avg.	Ord.	Gen.	Veri.	Pred.	Avg.	Anal.	Disc.	Avg.
Human Performance		17.5	15.9	61.3	55.5	37.6	69.8	31.2	45.6	21.8	42.1	–	–	–
Open-source MLLMs														
Qwen2.5-VL-7B-Instruct	×	32.0	33.9	49.0	62.4	42.6	56.2	20.8	20.7	1.3	24.6	25.2	21.4	23.3
MiMo-VL-7B-RL	×	34.2	33.7	44.2	62.4	42.4	43.9	28.5	18.5	11.4	27.4	28.7	25.9	27.3
MiMo-VL-7B-RL	✓	36.1	29.1	53.6	67.8	44.3	64.8	32.3	24.9	15.6	34.3	29.3	27.3	28.3
InternVL3–8B	×	27.5	31.0	38.8	65.6	39.4	43.4	20.4	20.2	3.9	23.9	29.2	25.3	27.2
InternVL3.5–8B	×	27.3	30.8	45.5	64.8	40.3	82.3	25.8	23.7	4.8	34.0	22.6	18.4	20.5
Intern-S1-mini	✓	33.3	31.2	52.5	61.4	42.5	73.6	14.3	16.8	8.3	28.1	33.5	28.3	30.9
Keye-VL-8B-Preview	✓	16.6	22.4	38.9	60.8	32.6	25.4	12.4	19.1	1.7	14.6	9.5	6.7	8.1
Keye–VL–1.5–8B	✓	21.0	23.4	51.3	64.0	37.0	56.7	9.5	20.0	2.8	22.1	8.4	6.1	7.2
GLM-4.1V-9B	✓	30.8	29.8	47.5	59.6	40.1	64.1	18.2	25.0	7.4	28.6	28.1	26.5	27.3
GLM-4.5V	✓	35.5	33.6	61.5	62.3	45.6	71.9	34.9	27.2	12.9	36.6	33.3	32.5	32.9
Kimi-VL-A3B-Thinking	✓	34.6	32.6	40.7	59.5	40.8	32.3	18.2	23.3	6.2	20.0	24.6	21.8	23.2
InternVL3.5–38B	✓	35.9	34.0	46.7	65.3	44.0	65.8	36.7	23.0	19.0	36.0	33.1	30.8	31.9
InternVL3–78B	✓	35.1	34.3	**73.2**	75.8	50.9	**87.1**	45.5	19.8	15.5	41.9	40.3	35.3	37.7
Qwen2.5-VL-72B-Instruct	×	30.5	34.7	54.5	64.5	43.9	86.3	34.1	23.8	0.3	35.9	31.9	29.3	30.6
Intern-S1	✓	38.9	35.2	58.9	73.8	49.9	82.2	45.0	24.1	15.4	36.0	43.0	36.3	39.6
Closed-source MLLMs														
Seed-VL-1.5	✓	32.9	24.6	43.9	69.2	40.7	73.9	48.6	19.8	27.9	42.5	32.0	29.4	30.7
Claude-Sonnet-4	×	25.6	31.2	54.3	61.9	40.8	78.7	37.6	16.5	11.6	36.0	29.1	30.1	29.6
Gemini-2.5-Flash	×	52.7	50.1	65.2	72.6	58.6	86.0	50.5	24.1	40.2	50.1	47.2	41.1	44.1
Gemini-2.5-Flash	✓	52.7	**50.7**	71.9	73.3	**60.2**	85.1	54.3	22.3	38.0	49.8	44.8	41.3	43.0
Gemini-2.5-Pro	×	**53.1**	45.9	64.3	**80.8**	59.2	83.7	61.3	**26.8**	49.6	53.8	50.6	45.2	47.9
Gemini-2.5-Pro	✓	51.3	44.3	63.8	74.4	56.7	84.2	59.9	**26.8**	46.9	54.3	50.1	44.8	47.4
GPT-5	✓	51.6	37.8	59.5	71.9	53.3	85.1	**66.9**	**26.8**	**51.8**	**57.5**	**55.4**	**57.4**	**56.4**

This suggests models can recognize correct sequences when presented with options but struggle to predict future steps independently.

4.3 Analysis and Insights

Effect of Chain-of-Thought Reasoning. The "thinking" mode shows mixed results across models and levels. Gemini–2.5–Flash benefits on Level–1 tasks (60.2% vs 58.6%) but slightly degrades on Level–2 (49.8% vs 50.1%), suggesting that extended reasoning may drift away from visual evidence in procedural tasks.

Critical Importance of Visual Information. Figure 3 demonstrates that removing video frames dramatically degrades performance across all levels, with Level-3 tasks showing near-complete failure without visual input. This validates our vision-centric design and confirms that scientific experiment understanding cannot rely solely on textual or prior knowledge.

Frame Number Analysis. Our analysis reveals that different models benefit differently from additional frames. Long-context models like Kimi-VL show continuous improvement up to 256 frames, while image-centric models saturate earlier. This suggests that architectural choices significantly impact the ability to process extended experimental procedures.

Domain-specific Challenges. Across the 13 scientific disciplines, we observe varying difficulty levels. Molecular biology and chemistry experiments show relatively higher performance due to standardized equipment and protocols, while materials science and neuroscience experiments prove more challenging due to specialized apparatus and subtle visual cues.

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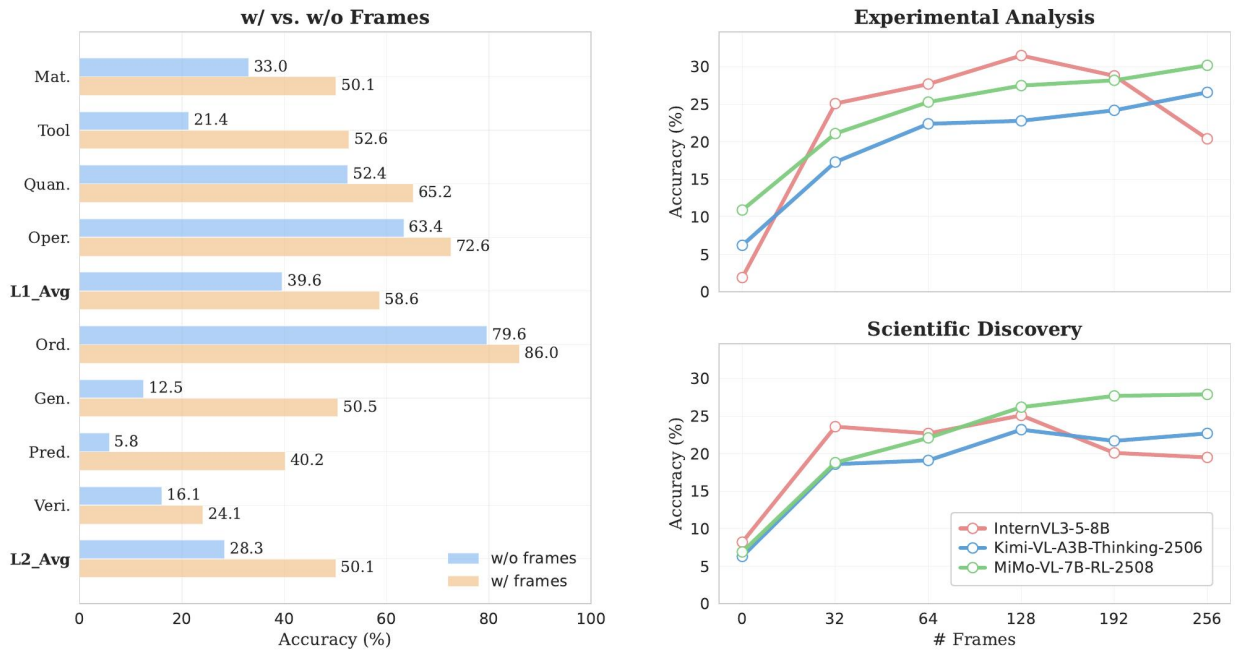

Figure 3: Effect of input video frames on model performance. Removing visual information severely degrades performance, particularly for Level-3 scientific reasoning tasks.

Error Analysis. Common failure modes include: (1) confusion between visually similar tools (e.g., different pipette types), (2) misunderstanding of procedural dependencies (e.g., thinking centrifugation can occur before sample preparation), and (3) inability to synthesize information across long experimental procedures for Level-3 reasoning.

4.4 Limitations and Future Directions

While ExpVidprovides a rigorous evaluation framework, several limitations warrant consideration. First, the reliance on JoVE videos, while ensuring quality and peer-review standards, may introduce domain-specific biases toward certain experimental styles or presentation formats. Second, our evaluation methodology, particularly LLM-based judging for Level-3 tasks, may introduce systematic biases despite careful validation.

Future work could expand ExpVidto include more diverse video sources, incorporate real-time experimental feedback scenarios, and develop more sophisticated evaluation metrics that capture nuanced scientific reasoning. Additionally, developing specialized MLLMs trained specifically for scientific content could push performance boundaries on these challenging tasks.

5 Conclusion

We introduce ExpVid, the first comprehensive benchmark specifically designed for evaluating multimodal large language models on authentic scientific experiment video understanding and reasoning. Through a carefully constructed three-level hierarchy spanning fine-grained perception, procedural understanding, and scientific reasoning, ExpVidprovides a rigorous evaluation framework that mirrors the complexity of real scientific experimentation.

Our benchmark comprises 7,800 expert-annotated question-answer pairs derived from 390 peer-reviewed laboratory videos across 13 scientific disciplines. The vision-centric design ensures that models cannot exploit textual shortcuts, requiring genuine visual comprehension of experimental procedures. Comprehensive evaluation of 19 state-of-the-art MLLMs reveals significant performance gaps, with the best models achieving only 57.5% accuracy on procedural understanding and 56.4% on scientific reasoning tasks.

Key findings from our evaluation include: (1) substantial challenges in fine-grained scientific visual recognition, where even human experts achieve only 37.6% accuracy; (2) asymmetric performance in procedural tasks, with models excelling at sequence recognition but struggling with step prediction; (3) critical importance of visual information, as

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performance collapses without video frames; and (4) inconsistent benefits of chain-of-thought reasoning, which may drift from visual evidence in complex scenarios.

ExpVidestablishes a challenging yet essential benchmark for the scientific community, highlighting critical gaps in current MLLM capabilities for scientific applications. As artificial intelligence increasingly supports scientific research and education, the ability to understand and reason about experimental procedures becomes paramount. Our benchmark provides a concrete roadmap for developing MLLMs capable of serving as effective scientific assistants, educational tools, and research aids.

The significant performance gaps revealed by our evaluation underscore the need for continued research in scientific video understanding. Future developments in specialized training methodologies, architectural innovations for long-form video processing, and enhanced reasoning capabilities will be essential for building MLLMs that can truly understand the nuanced world of scientific experimentation.

We hope ExpVidwill catalyze research in scientific video understanding and contribute to the development of more capable, reliable MLLMs for the scientific community. The benchmark, along with all associated data and evaluation scripts, will be made publicly available to support reproducible research and continued progress in this important domain.

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